

Each question is worth 6 points for a total of 30 points. The extra Credit can get you an additional 5 points making the highest possible score 35 out of 30. Write solutions using COMPLETE sentences.

Problem 1

Let $A = \{1, 2\}$ and $B = \{3, 4, 5\}$. Write down all functions $f : A \rightarrow B$. Indicate which are one-to-one and which are onto B .

Problem 2

For each of the following functions, find $f(2)$.

a) $f = \{(x, y) : x, y \in \mathbb{Z}, x + y = 0\}$

b) $f = \{(1, 2), (2, 3), (3, 2)\}$

c) $f = \{1, 2, 3, 4\} \times \{1\}$

d) $f : \mathbb{N} \rightarrow \mathbb{N}$ where $f(n) = n!$

Problem 3

Let A and B be sets, and let $f : A \rightarrow B$. Prove $f \circ id_A = id_B \circ f = f$.

Problem 4

Let A and B be sets and suppose $f : A \rightarrow B$ is one-to-one and onto. Prove $f \circ f^{-1} = id_B$ and $f^{-1} \circ f = id_A$.

Problem 5

Please express $\pi = (1, 2) \circ (2, 3) \circ (3, 4) \circ (4, 5) \circ (5, 1)$ in disjoint cycle form.

Extra Credit:

Let $f : \mathbb{Z} \rightarrow \mathbb{N}$ by $f(x) = |x|$. Determine if f is one-to-one, if it is onto, and prove your answers.